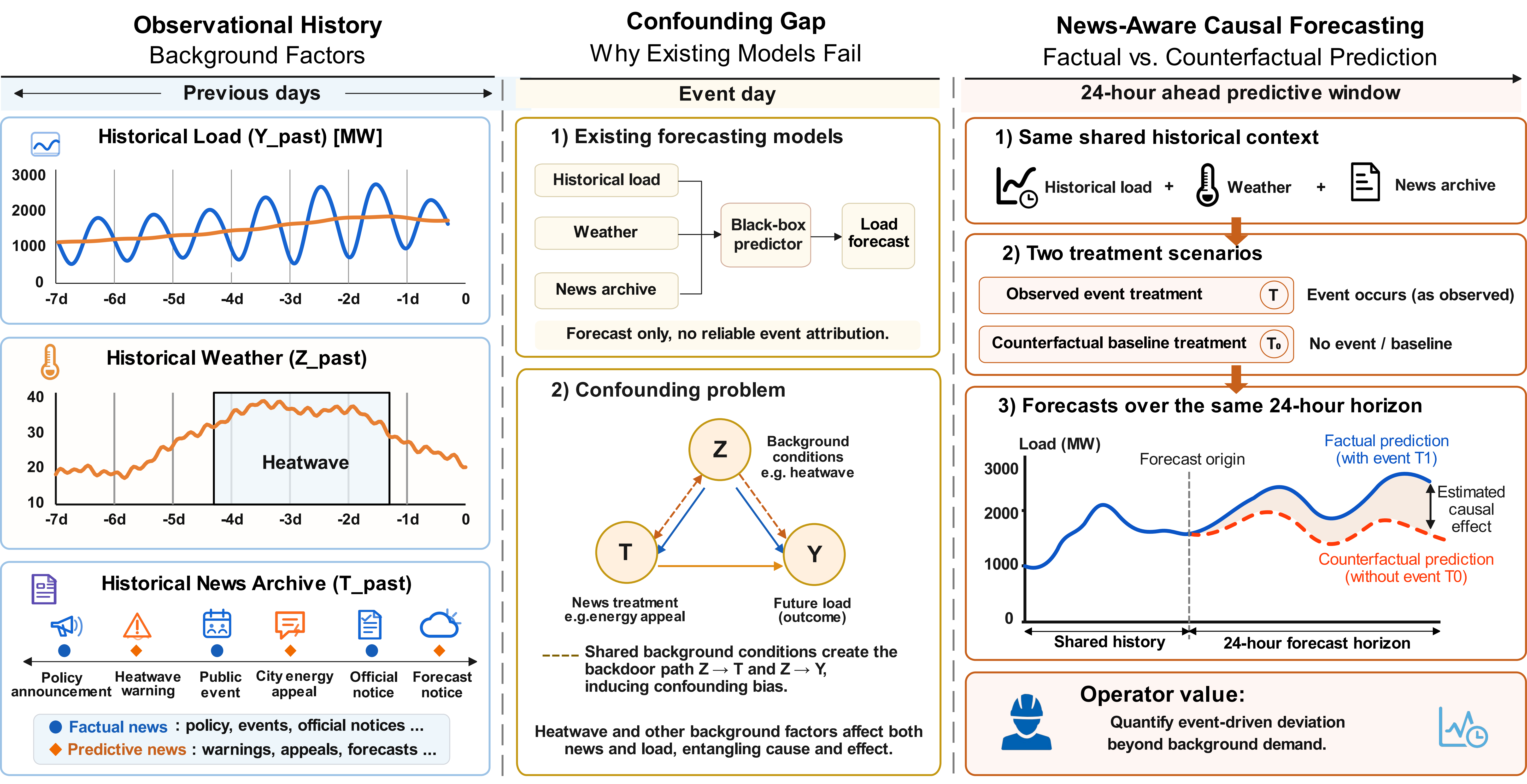
Observational History
Background Factors
Previous days
Historical Load (Y_past) [MW]
3000
2000
1000
0
-7d
-6d
-5d
-4d
-3d
-2d
-1d
0
Historical Weather (Z_past)
40
30
20
10
Heatwave
-7d
-6d
-5d
-4d
-3d
-2d
-1d
0
Historical News Archive (T_past)
Policy announcement
Heatwave warning
Public event
City energy appeal
Official notice
Forecast notice
Factual news : policy, events, official notices ...
Predictive news : warnings, appeals, forecasts ...
Confounding Gap
Why Existing Models Fail
Event day
1) Existing forecasting models
Historical load
Weather
News archive
Black-box predictor
Load forecast
Forecast only, no reliable event attribution.
2) Confounding problem
Z
Background conditions e.g. heatwave
T
Y
News treatment e.g.energy appeal
Future load (outcome)
Shared background conditions create the backdoor path Z → T and Z → Y, inducing confounding bias.
Heatwave and other background factors affect both news and load, entangling cause and effect.
News-Aware Causal Forecasting
Factual vs. Counterfactual Prediction
24-hour ahead predictive window
1) Same shared historical context
Historical load
+
Weather
+
News archive
2) Two treatment scenarios
Observed event treatment
T
Event occurs (as observed)
Counterfactual baseline treatment
T0
No event / baseline
3) Forecasts over the same 24-hour horizon
Load (MW)
3000
2000
1000
0
Forecast origin
Factual prediction (with event T1)
Estimated causal effect
Counterfactual prediction (without event T0)
Shared history
24-hour forecast horizon
Operator value:
Quantify event-driven deviation beyond background demand.